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TIME-DIVISION DUPLEX (TDD) ANTENNA SYSTEM

Abstract

One example includes a self-synchronizing TDD antenna system. The system includes an antenna to communicate transmit and receive signals and an antenna circuit coupled to a user communication system via a transmission line cable. The antenna circuit includes a transmission line measurement circuit to determine signal loss through the transmission line cable and an amplitude adjustment circuit to adjust amplitude of the transmit and/or receive signals based on the determined signal loss. The antenna circuit also includes a transmit detection circuit to monitor signal power of the transmit signal, and a controller to switch the amplitude adjustment circuit from a receive mode to a transmit mode in response to the monitored signal power exceeding a predetermined threshold. In the receive mode, the adjustment circuit applies a receive amplitude adjustment to the receive signal, and in the transmit mode the adjustment circuit applies a transmit amplitude adjustment to the transmit signal.

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Background/Summary

[0001] This application is a continuation of U.S. patent application Ser. No. 17/278,531 filed Mar. 22, 2021, entitled “TIME-DIVISION DUPLEX (TDD) ANTENNA SYSTEM”, which is a national stage entry under 35 U.S.C. § 371 of International Application No. PCT/US2019/054555, filed on Oct. 3, 2019, entitled “TIME-DIVISION DUPLEX (TDD) ANTENNA SYSTEM”, which claims priority to, and the benefit of, U.S. Provisional Application No. 62/741,311, filed Oct. 4, 2018, entitled “SELF-SYNCHRONIZING TIME DIVISION DUPLEX ANTENNA”, and U.S. Provisional Application No. 62/768,026, filed Nov. 15, 2018 entitled “SELF-CALIBRATING TIME DIVISION DUPLEX (TDD) DUAL-POLARIZED ANTENNA”. The foregoing application is hereby incorporated by reference in its entirety (except for any subject matter disclaimers or disavowals, and except to the extent of any conflict with the disclosure of the present application, in which case the disclosure of the present application shall control).

TECHNICAL FIELD

[0002] This disclosure relates generally to communication systems, and more specifically to a time-division duplex (TDD) antenna system.

BACKGROUND

[0003] An antenna array (or array antenna) is a set of multiple antenna elements that work together as a single antenna to transmit or receive radio waves. The individual antenna elements can be connected to a receiver and/or transmitter by circuitry that applies an appropriate amplitude and/or phase adjustment of signals received and/or transmitted by the antenna elements. When used for transmitting, the radio waves radiated by each individual antenna element combine and superpose with each other, adding together (interfering constructively) to enhance the power radiated in desired directions, and cancelling (interfering destructively) to reduce the power radiated in other directions. Similarly, when used for receiving, the separate received signals from the individual antenna elements are combined with the appropriate amplitude and/or phase relationship to enhance signals received from the desired directions and cancel signals from undesired directions.

SUMMARY

[0004] One example includes a self-synchronizing TDD antenna system. The system includes an antenna to communicate transmit and receive signals and an antenna circuit coupled to a user communication system via a transmission line cable. The antenna circuit includes a transmission line measurement circuit to determine signal loss through the transmission line cable and an amplitude adjustment circuit to adjust amplitude of the transmit and/or receive signals based on the determined signal loss. The antenna circuit also includes a transmit detection circuit to monitor signal power of the transmit signal, and a controller to switch the amplitude adjustment circuit from a receive mode to a transmit mode in response to the monitored signal power exceeding a predetermined threshold. In the receive mode, the adjustment circuit applies a receive amplitude adjustment to the receive signal, and in the transmit mode the adjustment circuit applies a transmit amplitude adjustment to the transmit signal.

[0005] Another example includes a method for communicating at least one of a transmit signal and a receive signal via a time-division duplex (TDD) antenna system comprising an antenna. The method includes providing a calibration signal from an antenna circuit associated with the antenna system to a user communication system on at least one transmission line cable. The method also includes receiving a returned signal corresponding to the calibration signal retransmitted back from the user communication system to the antenna circuit and determining signal loss between the user communication system and the antenna circuit through at least one transmission line cable based on the returned signal. The method also includes monitoring signal power of the transmit signal obtained from the user communication system via the at least one transmission line cable and adjusting an amplitude of the receive signal in a receive mode based on the determined signal loss. The method further includes switching an amplitude adjustment circuit from the receive mode to a transmit mode in response to the monitored signal power exceeding a predetermined threshold, and adjusting an amplitude of the transmit signal in the transmit mode based on the determined signal loss.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates an example of a communication system.

[0007] FIG. **2** illustrates an example of an antenna system.

[0008] FIG. **3** illustrates an example diagram of calibration of a communication system.

[0009] FIG. **4** illustrates another example diagram of calibration of a communication system.

[0010] FIG. **5** illustrates an example of a calibration circuit.

[0011] FIG. **6** illustrates another example of a calibration circuit.

[0012] FIG. **7** illustrates an example of a transmission line measurement circuit.

[0013] FIG. **8** illustrates an example of a controller.

[0014] FIG. **9** illustrates an example of a TDD communication stream.

[0015] FIG. **10** illustrates an example of an amplitude adjustment circuit.

[0016] FIG. **11** illustrates another example of an amplitude adjustment circuit.

[0017] FIG. **12** illustrates another example of an amplitude adjustment circuit.

[0018] FIG. **13** illustrates another example of an amplitude adjustment circuit.

[0019] FIG. **14** illustrates another example of an amplitude adjustment circuit.

[0020] FIG. **15** illustrates another example of an amplitude adjustment circuit.

[0021] FIG. **16** illustrates an example of a method for communicating at least one of a transmit signal and a receive signal via a time-division duplex (TDD) antenna system comprising an antenna.

DETAILED DESCRIPTION

[0022] This disclosure relates generally to communication systems, and more specifically to a time-division duplex (TDD) antenna system. A communication system can be implemented that includes a user communication system and an antenna system. As an example, the communication system can be implemented as a wireless broadband communication system, such as using a Long Term Evolution (LTE) communication standard. The antenna system can be physically communicatively coupled (e.g., via a transmission line cable) to the user communication system to provide enhanced wireless communication capability for the communication system, such as to provide wireless extension or capability of the user communication system to communicate with a base station (e.g., in a time-division duplex (TDD) manner). For example, the antenna system can provide wireless communication capability for the user communication system based on the user communication system being located in a location that prohibits or impedes wireless connection to a base station based on, for example, intervening physical barriers or extreme range.

[0023] The antenna system includes one or more antenna arrays and an antenna circuit. The antenna array(s) can be arranged as any of a variety of antenna arrays to provide a respective one or more wireless signals to be transmitted from and/or received at the antenna system. For example, the antenna array(s) can include an arrangement of antenna elements (e.g., strip-line conductors) to provide signal diversity between two or more respective signal paths, such as based on polarization diversity (e.g., orthogonal polarizations of two separate signal paths). The antenna array(s) can thus each transmit signal(s) and receive signal(s), such as in a TDD manner based on a defined standard on which the user communication system operates. As described herein, the antenna circuit can determine cable losses of the interconnection between the antenna system and the user communication system, such as in a calibration procedure. As a result, the antenna circuit can be implemented to provide attenuation of transmitted signals (hereinafter "transmit signals") provided from the user communication system via the antenna array(s) in a manner that allows the transmit signals to be transmitted at or below a predetermined effective isotropic radiated power (EIRP), such as defined by the operating standard of the user communication system. Additionally, as also described herein, the antenna circuit can be configured to monitor signal power on the respective communication paths to facilitate the TDD operation of the communication system without any input from the user communication system. As a result, the antenna system can be installed to cooperate with the user communication system in a manner that is substantially agnostic of the interconnection between the antenna system and the user communication system, and without active communication between the antenna system and the user communication system.

[0024] FIG. **1** illustrates an example of a communication system **10**. The communication system **10** can be implemented as a wireless broadband communication system, such as using a Long Term Evolution (LTE) communication standard. In the example of FIG. **1**, the communication system **10** includes a user communication system **12** and an antenna system **14**. As an example, the user communication system **12** can correspond to a wireless gateway, such as to facilitate wireless communications (e.g., Wi-Fi, Bluetooth, and/or cellular communication) between one or more user devices and a wireless network, such as a cellular network or other wide-area network (WAN).

[0025] In the example of FIG. **1**, the antenna system **14** is communicatively coupled to a calibration circuit **15** via at least one transmission line cable **16** (e.g., an RG6 cable), such that the calibration circuit **15** interconnects the user communication system **12** and the antenna system **14**. As an example, the calibration circuit **15** can include at least one bias-tee configured to inject a DC voltage onto the transmission line cable(s) **16** that are provided to the antenna system **14**. Alternatively, the user communication system **12** can provide the DC voltage injection onto the transmission line cable(s) **16**. As a result, the DC voltage can provide input power to the antenna system **14**, as described in greater detail herein.

[0026] As an example, the antenna system **14** can provide enhanced wireless communication capability for the user communication system **12**, such as to provide wireless extension or capability of the user communication system **12** to communicate with a base station (e.g., in a time-division duplex (TDD) manner). For example, the antenna system **14** can provide wireless communication capability for the user communication system **12** based on the user communication system **12** being located in a location that prohibits or impedes wireless connection to a base station based on, for example, intervening physical barriers or extreme range.

[0027] The antenna system **14** includes one or more antenna arrays **18** and an antenna circuit **20**. The antenna array(s) **18** can be arranged as any of a variety of antenna arrays to provide a respective one or more wireless signals to be transmitted from and/or received at the antenna system **14**. For example, the antenna array(s) **18** can include an arrangement of antenna elements (e.g., strip-line conductors) to provide signal diversity between two or more respective signal paths, such as based on polarization diversity. For example, the antenna array(s) **18** can include two separate arrays of orthogonally polarized antenna elements to provide orthogonal polarizations of signals propagating in two separate respective signal paths through the antenna circuit **20**. The antenna array(s) **18** can thus each transmit signal(s) and receive signal(s), such as bidirectionally in a TDD manner based on a defined standard on which the user communication system **12** operates. In the example of FIG. **1**, signals transmitted from and received at the antenna array(s) **18** are demonstrated as signals “RF”, and the same signals propagating bidirectionally along the transmission line cable(s) **16** are demonstrated as signals “TS”.

[0028] The antenna circuit **20** includes a transmission line measurement circuit **22** and an amplitude adjustment circuit **24**. The transmission line measurement circuit **22** is configured to determine a signal loss between the user communication system **12** (e.g., the calibration circuit **15**) and the antenna circuit **14** through the at least one transmission line cable **16**. For example, during installation of the antenna system **14** and/or periodically thereafter, the transmission line measurement circuit **22** can initiate a calibration operation (e.g., in response to a calibration command). As an example, during the calibration operation, the transmission line measurement circuit **22** can be configured to generate a calibration signal, such as a radio frequency (RF) signal, that can be transmitted to the user communication system **12** from the antenna system **14** via the transmission line cable(s) **16**, such that the calibration signal can be retransmitted back to the antenna system **14** from the user communication system **12** via the transmission line cable(s) **16**. As a result, the transmission line measurement circuit **22** can measure at least one characteristic of the return signal (e.g., power) to determine signal loss exhibited by the transmission line cable(s) **16**.

[0029] In response to determining the signal loss, the amplitude adjustment circuit **24** can be configured to adjust an amplitude of at least one of transmit signals transmitted from the antenna system **14** and receive signals received at the antenna system **14** based on the determined signal loss. As described herein, the term “transmit signals” refers to signals that are originated at the user communication system **12**, propagate through the transmission line cable(s) **16** as a signal TS, and are transmitted from the antenna system **14** via the antenna array(s) **18** as a signal RF. Similarly, the term “receive signals” refers to signals that are received at the antenna system **14** via the antenna array(s) **18** as a signal RF, propagate through the transmission line cable(s) **16** as a signal TS, and are provided to the user communication system **12**. The amplitude adjustment circuit **24** can therefore adjust the amplitude of the transmit and receive signals in separate respective signal paths in the antenna circuit **20** based on the signal loss determined during the calibration operation.

[0030] For example, the communication system **10** can be configured to operate based on a predetermined communication standard that can dictate a predetermined maximum effective isotropic radiated power (EIRP), such as +23 dBm for the transmit signals. As an example, the amplitude adjustment circuit **24** can include one or more variable circuit elements (VCEs) to amplify or attenuate the transmit signals (e.g., down to less than the predetermined maximum EIRP) and/or the receive signals (e.g., down to less than a maximum saturation power associated with the antenna circuit **14** and/or the user communication system **12**). For example, the antenna array(s) **18** can be designed with sufficient gain, or the antenna circuit **20** can be sufficiently high to provide the transmit signals at a power level that is greater than the predetermined maximum EIRP (e.g., to overcome power losses of the transmission line cable(s) **16** regardless of the length of the transmission line cable(s) **16**), such that the transmit signals can be attenuated down to approximately the predetermined maximum EIRP. Therefore, the antenna system **14** can be installed in a manner that is substantially agnostic of the length and/or loss characteristics of the transmission line cable(s) **16** based on the calibration operation to determine the signal loss of the transmission line cable(s) **16**.

[0031] In the example of FIG. **1**, the antenna circuit **20** also includes a transmit detection circuit **26** and a controller **28**. As described previously, the communication system **10** can operate based on a TDD communication standard, such that the transmit signals and the receive signals can be interleaved with each other on a given signal path between the user communication system **12** and the antenna array(s) **18**. The transmit detection circuit **26** can be configured to measure power on a given signal path in the antenna circuit **20** to determine if user communication system **12** is transmitting a transmit signal. Therefore, in response to determining if the user communication system **12** is transmitting a transmit signal, the controller **28** can switch the adjustment circuit **24** from a receive mode (e.g., as a default mode) to a transmit mode to facilitate transmission of the transmit signal from the antenna system **14** via the antenna array(s) **18**. Additionally, in response to the transmit detection circuit **26** detecting a decrease in the power of the signal path (e.g., less than the predetermined threshold), the controller **28** can switch the adjustment circuit **24** back to the receive mode from the transmit mode (e.g., upon expiration of a timer).

[0032] In response to the transmit detection circuit **26** determining that the user communication system **12** is transmitting a transmit signal, such as based on the power on the signal path being greater than a predetermined threshold, the controller **28** can provide a signal to the amplitude adjustment circuit **24** to switch the signal path from a receive mode to a transmit mode. Therefore, the amplitude adjustment circuit **24** can provide the appropriate amplitude adjustment to the transmit signal (e.g., via a VCE) to facilitate transmission of the transmit signal from the antenna system **14** via the antenna array(s) **18**. As an example, the amplitude adjustment circuit **24** can include power amplifier and/or a filter in each of the transmit and receive switchable portions of the signal path, and/or can include a short-circuit bypass path in one of the transmit and receive signal paths.

[0033] As a result, the antenna system **14** can operate to facilitate the bidirectional TDD communications between transmit and receive signals without requiring communication or signal transfer from the user communication system **12**. Therefore, the antenna system **14** can be installed in a simplistic manner that is largely independent of the operation of the user communication system **12**. Additionally, as previously described, the antenna system **14** can be installed in a manner that is agnostic of the length of the transmission line cable(s) **16** interconnecting the antenna system **14** and the user communication system **12**. Accordingly, and as described in greater detail herein, the antenna system **14** can be simplistically installed to efficiently facilitate wireless communication between the user communication system **12** and a network hub (e.g., a base station).

[0034] FIG. **2** illustrates an example of an antenna system **50**. The antenna system **50** can correspond to the antenna system **14** in the example of FIG. **1**. Therefore, reference is to be made to the example of FIG. **1** in the following description of the example of FIG. **2**.

[0035] The antenna system **50** includes an antenna circuit **52** that is demonstrated in the example of FIG. **2** as including a first signal path **54** and a second signal path **56** that can each correspond to a separate signal diversity type, as described in greater detail herein. The antenna system **50** also includes a first antenna array **58** and a second antenna array **60** that can each be associated with the respective signal diversity types. The antenna arrays **58** and **60** can be arranged as any of a variety of antenna arrays to provide a respective one or more wireless signals to be transmitted from and/or received at the antenna system **50**, demonstrated as respective signals RF.sub.1 and RF.sub.2. For example, the antenna arrays **58** and **60** can include an arrangement of antenna elements (e.g., strip-line conductors) to provide the signal diversity between the two respective signal paths, such as based on polarization diversity. As an example, the antenna arrays **58** and **60** can be configured as separate respective arrays of orthogonally polarized antenna elements to provide orthogonal polarizations of signals propagating in the respective signal paths. The antenna arrays **58** and **60** can thus each transmit and receive signals a TDD manner based on a defined standard on which the user communication system (not shown in the example of FIG. **2**) operates.

[0036] Additionally, the antenna system **50** is communicatively coupled to the user communication system (e.g., the user communication system **12**) via a first transmission line cable **62** configured to propagate a signal TS.sub.1 between the user communication system and the antenna circuit **52** and a second transmission line cable **64** configured to propagate a signal TS.sub.2 between the user communication system and the antenna circuit **52**. For example, the transmission line cables **62** and **64** can be connected to a calibration circuit (e.g., the calibration circuit **15**) that is coupled to the user communication system. The transmission line cables **62** and **64** can each be associated with the respective signal diversity types, and thus the respective signal paths of the antenna circuit **52**. For example, the transmission line cables **62** and **64** can be configured as RG6 cables or other types of transmission line cables. In the example of FIG. **2**, the antenna circuit **52** includes an extraction circuit **66** that can be configured as a DC decoupler (e.g., a bias-tee) that is coupled to the transmission line cables **62** and **64** to extract a DC voltage, demonstrated as a voltage V.sub.DC, that is provided, such as from the calibration circuit **15**, to power the electronics of the antenna circuit **52**. Therefore, the antenna system **50** does not require a local power source, such that the antenna system **50** can be installed in a more flexible manner.

[0037] The antenna circuit **52** also includes a transmission line measurement circuit **68**. The transmission line measurement circuit **68** is configured to determine a signal loss between the user communication system and the antenna circuit **50** through the transmission line cables **62** and **64**. For example, during installation of the antenna system **50** and/or periodically thereafter, the transmission line measurement circuit **68** can initiate a calibration operation in response to a calibration command CAL. As an example, the calibration command CAL can be provided in response to a user input, such as via a physical input on the antenna system **50** or the calibration system **15** (e.g., a physical button or a button on a touchscreen), in response to power-up of the antenna system **50** (e.g., automatically in response to initially receiving the voltage V.sub.DC), periodically from a processor or controller device (e.g., at periodic or programmable intervals), or from any of a variety of other means.

[0038] In the example of FIG. **2**, the transmission line measurement circuit **68** includes a calibration signal generator **70**, a signal monitor **72**, and a memory **74**. The transmission line measurement circuit **68** is communicatively coupled to the first transmission line cable **62** (via the extraction circuit **66**) through a first switch SW.sub.1 and to the second transmission line cable **64** (via the extraction circuit **66**) through a second switch SW.sub.2. In the example of FIG. **2**, the switches SW.sub.1 and SW.sub.2 are demonstrated as being set to a normal operating mode state, and are controlled via the calibration command CAL. Therefore, in the normal operating mode, as in the state demonstrated in the example of FIG. **2**, the switch SW.sub.1 connects the first signal path **54** to the first transmission line cable **62** and the switch SW.sub.2 connects the second signal path **54** to the second transmission line cable **64** to facilitate propagation of the transmit and receive signals between the user communication system and the antenna arrays **58** and **56** via the respective signal paths **54** and **56** and the respective transmission line cables **62** and **64**. However, in a calibration mode, such as initiated by the calibration command CAL, the switches SW.sub.1 and SW.sub.2 can be switched to coupling the transmission line measurement circuit **68** to the first transmission line cable **62** via the first switch SW.sub.1 and to the second transmission line cable **64** via the second switch SW.sub.2 to facilitate the calibration operation.

[0039] During the calibration operation, the calibration signal generator **70** can be configured to generate a calibration signal, such as a dummy RF signal having a predefined frequency. In the example of FIG. **2**, the calibration signal is demonstrated as a first calibration signal CS.sub.1 and a second calibration signal CS.sub.2 corresponding, respectively, to the first and second transmission line cables **62** and **64**. The transmission line measurement circuit **68** can thus transmit one or both of the calibration signals CS.sub.1 and CS.sub.2 to the user communication system via the respective one of the transmission line cables **62** and **64**.

[0040] The user communication system can be configured to retransmit the calibration signal(s) CS.sub.1 and/or CS.sub.2 back to the antenna system **50** from the user communication system as respective return signal(s) via the transmission line cables **62** and **64** during the calibration procedure. In response to receiving the return signal(s), the signal monitor **72** can be configured to measure at least one characteristic of the return signal(s) to determine signal loss exhibited by the transmission line cables **62** and **64**. For example, the characteristic of the return signal can be power, such that the signal monitor **72** can calculate a power ratio between one of the calibration signals CS.sub.1 and CS.sub.2 and the respective return signal. Accordingly, the signal monitor **72** can determine the signal loss exhibited by the transmission line cable(s) **62** and **64** based on the power ratio between the calibration signal(s) CS.sub.1 and CS.sub.2 and the respective return signal. Additionally, other types of characteristics can be monitored instead of or in addition to power, such as delay time, to determine the signal loss exhibited by the transmission line cable(s) **62** and **64**. The transmission line measurement circuit **68** can then store the determined signal loss in the memory **74**. For example, the memory **74** can store signal loss information for both of the transmission line cables **62** and **64** individually or in combination, and can store the signal loss information for each calibration operation or for the most recent calibration operation. As an example, the memory **74** can be configured as a non-volatile memory, such that the memory **74** can retain the calculated signal loss information during power loss of the antenna system **50**. As a result, the calculated signal loss can be retrieved from the memory **74** after power is returned to the antenna system **50** to facilitate operation of the antenna system **50** without the need for a calibration operation.

[0041] FIGS. **3** and **4** illustrate example diagrams **100** and **150**, respectively, of calibration of a communication system. In the example of FIG. **3**, the communication system includes a calibration circuit **102** and an antenna system **104** that are communicatively coupled by a first transmission line cable **106** and a second transmission line cable **108**. In the example of FIG. **4**, the communication system includes a calibration circuit **152** and an antenna system **154** that are communicatively coupled by a first transmission line cable **156** and a second transmission line cable **158**. The calibration circuits in the examples of FIGS. **3** and **4** can each correspond to the calibration circuit **15** in the example of FIG. **1**. As an example, the calibration circuits **102** and **152** can correspond to the calibration circuit **15**, the antenna systems **104** and **154** can correspond to the antenna systems **14** and/or **50**, and the transmission line cables **106** and **108** and the transmission line cables **156** and **158** can correspond to the transmission line cable(s) **16** an/or the transmission line cables **62** and **64**. Therefore, reference is to be made to the examples of FIGS. **1** and **2** in the following description of the examples of FIGS. **3** and **4**.

[0042] In the diagram **100**, the antenna circuit **104** includes a transmission line measurement circuit **110**. During the calibration operation, the transmission line measurement circuit **110** can be configured (e.g., via the calibration signal generator **70**) to generate a first calibration signal CS.sub.1 that is transmitted by the transmission line measurement circuit **110** along the first transmission line cable **106**. The calibration circuit **102** can be configured to retransmit (e.g., reflect) the calibration signal CS.sub.1 back to the antenna system **104** as a respective return signal RTN.sub.1 via the first transmission line cable **106**. Therefore, in the diagram **100**, the first transmission line **106** is configured to propagate both the calibration signal CS.sub.1 and the reflected return signal RTN.sub.1. In response to receiving the return signal RTN.sub.1, the transmission line measurement circuit **110** can be configured (e.g., via the signal monitor **72**) to measure the at least one characteristic of the reflected return signal RTN.sub.1 to determine the signal loss exhibited by the first transmission line cable **106**. For example, the characteristic of the reflected return signal RTN.sub.1 can be power, such that the transmission line measurement circuit **110** can calculate a power ratio between the calibration signal CS.sub.1 and the respective reflected return signal RTN.sub.1 to determine the signal loss exhibited by the first transmission line cable **106**.

[0043] Similarly, the transmission line measurement circuit **110** can repeat the previously described calibration procedure with respect to the second transmission line cable **108**. For example, the transmission line measurement circuit **110** can also be configured to generate a second calibration signal CS.sub.2 that is transmitted by the transmission line measurement circuit **110** along the second transmission line cable **108**. The calibration circuit **102** can be configured to retransmit (e.g., reflect) the calibration signal CS.sub.2 back to the antenna system **104** as a respective return signal RTN.sub.2 via the second transmission line cable **108**. In response to receiving the reflected

return signal RTN.sub.2, the transmission line measurement circuit **110** can be configured (e.g., via the signal monitor **72**) to measure the at least one characteristic of the reflected return signal RTN.sub.2 to determine the signal loss exhibited by the second transmission line cable **108**, similar to as described previously with respect to the first transmission line cable **106**.
[0044] In the diagram **150**, the antenna circuit **154** includes a transmission line measurement circuit **160**. During the calibration operation, the transmission line measurement circuit **160** can be configured (e.g., via the calibration signal generator **70**) to generate a first calibration signal CS.sub.1 that is transmitted by the transmission line measurement circuit **160** along the first transmission line cable **156**. The calibration circuit **152** can be configured to retransmit the calibration signal CS.sub.1 back to the antenna system **154** as a respective return signal RTN.sub.1 via the second transmission line cable **158**. Therefore, in the diagram **150**, the first transmission line **156** is configured to propagate the calibration signal CS.sub.1 and the second transmission line cable **158** is configured to propagate the return signal RTN.sub.1. In response to receiving the return signal RTN.sub.1, the transmission line measurement circuit **160** can be configured (e.g., via the signal monitor **72**) to measure the at least one characteristic of the return signal RTN.sub.1 to determine the signal loss exhibited by the first and second transmission line cables **156** and **158**. For example, the characteristic of the return signal RTN.sub.1 can be power, such that the transmission line measurement circuit **160** can calculate a power ratio between the calibration signal CS.sub.1 and the respective return signal RTN.sub.1 to determine the signal loss exhibited by the first and second transmission line cables **156** and **158**. As an example, the transmission line measurement circuit **160** can either conclude the calibration operation, or can repeat the previously described calibration procedure with respect to switching the first and second transmission line cables **156** and **158** with respect to transmission of the calibration signal CS and receipt of the return signal RTN.
[0045] FIG. **5** illustrates an example of a calibration circuit **200**. The calibration circuit **200** can correspond to the calibration circuit **15** of the example of FIG. **1**. Therefore, reference is to be made to the example of FIG. **1** in the following description of the example of FIG. **5**. The calibration circuit **200** is configured to connect to the user communication system (e.g., the user communication system **12**) to provide DC power to the antenna system **50**. For example, the DC power can be a DC bias voltage that is provided on the transmission line cables **62** and **64**, such that the extraction circuit **66** extracts the voltage V.sub.DC to provide power to the electronic components of the antenna system **50**.
[0046] The calibration circuit **200** includes a power block **202** that is configured to generate or receive the DC voltage V.sub.DC (e.g., approximately 3.3 volts). For example, the power block **202** can be configured as a low-dropout (LDO) voltage regulator, such as to generate the DC voltage V.sub.DC from a higher input voltage (e.g., approximately 5 volts). The power block **202** provides the DC voltage V.sub.DC to a first injection circuit **204** and a second injection circuit **206**. For example, the injection circuits **204** and **206** can be configured as bias-tees. The injection circuits **204** and **206** are each coupled to the transmission line cables **62** and **64** on which the transmit and receive signals TS.sub.1 and TS.sub.2, respectively, are propagated. Therefore, the injection circuits **204** and **206** are configured to provide the DC voltage V.sub.DC onto the transmission line cables **62** and **64**.
[0047] The calibration circuit **200** includes a first signal port **208** and a second signal port **210** that are coupled to the user communication system **12**. The first signal port **208** is configured to propagate the transmit and receive signals TS.sub.1 and the second signal port **210** is configured to propagate the transmit and receive signals TS.sub.2. The first signal port **208** is coupled to the first transmission line cable **62** via a first switch SW.sub.3 and the second signal port **210** is coupled to the second transmission line cable **64** via a second switch SW.sub.4. In the example of FIG. **5**, the switches SW.sub.3 and SW.sub.4 are demonstrated as being set to a normal operating mode state, and are controlled via the calibration command CAL. For example, the calibration command CAL

can correspond to the same calibration command CAL described in the example of FIG. **2**, or can be a different calibration command (e.g., facilitated by a user) that is to be provided during the calibration operation.

[0048] Therefore, in the normal operating mode, as in the state demonstrated in the example of FIG. **5**, the switch SW.sub.3 connects the first signal port **208** to the first transmission line cable **62** and the switch SW.sub.4 connects the second signal port **210** to the second transmission line cable **64** to facilitate propagation of the transmit and receive signals between the user communication system and the antenna system **50** via the respective transmission line cables **62** and **64**. However, in the calibration mode, such as initiated by the calibration command CAL, the switches SW.sub.3 and SW.sub.4 can be switched to provide a short circuit between the transmission line cables **62** and **64**. As a result, the calibration signal CS.sub.1, such as provided in the example of FIG. **4**, can be provided from the first transmission line cable **62** and can be retransmitted back to the antenna system **50** as the return signal RTN.sub.1 along the second transmission line cable **64**. Therefore, with minimal input, the calibration circuit **200** can be implemented in the calibration operation to determine the signal loss of the transmission line cables **62** and **64**.

[0049] FIG. **6** illustrates an example of a calibration circuit **230**. The calibration circuit **230** can correspond to the calibration circuit **15** of the example of FIG. **1**. Therefore, reference is to be made to the example of FIG. **1** in the following description of the example of FIG. **6**. The calibration circuit **230** is configured to connect to the user communication system (e.g., the user communication system **12**) to provide DC power to the antenna system **50**. For example, the DC power can be a DC bias voltage that is provided on the transmission line cables **62** and **64**, such that the extraction circuit **66** extracts the voltage V.sub.DC to provide power to the electronic components of the antenna system **50**.

[0050] The calibration circuit **230** includes a power block **232** that is configured to generate or receive the DC voltage V.sub.DC (e.g., approximately 3.3 volts). For example, the power block **232** can be configured as an LDO voltage regulator, such as to generate the DC voltage V.sub.DC from a higher input voltage (e.g., approximately 5 volts). The power block **232** provides the DC voltage V.sub.DC to a first injection circuit **234** and a second injection circuit **236**. For example, the injection circuits **234** and **236** can be configured as bias-tees. The injection circuits **234** and **236** are each coupled to the transmission line cables **62** and **64** on which the transmit and receive signals TS.sub.1 and TS.sub.2, respectively, are propagated. Therefore, the injection circuits **234** and **236** are configured to provide the DC voltage V.sub.DC onto the transmission line cables **62** and **64**.

[0051] The calibration circuit **230** includes a first signal port **238** and a second signal port **240** that are coupled to the user communication system **12**. The first signal port **238** is configured to propagate the transmit and receive signals TS.sub.1 and the second signal port **240** is configured to propagate the transmit and receive signals TS.sub.2. The first signal port **238** is coupled to the first transmission line cable **62** via a first switch SW.sub.3 and the second signal port **240** is coupled to the second transmission line cable **64** via a second switch SW.sub.4. In the example of FIG. **6**, the switches SW.sub.3 and SW.sub.4 are demonstrated as being set to a normal operating mode state, and are controlled via the calibration command CAL. For example, the calibration command CAL can correspond to the same calibration command CAL described in the example of FIG. **2**, or can be a different calibration command (e.g., facilitated by a user) that is to be provided during the calibration operation.

[0052] Therefore, in the normal operating mode, as in the state demonstrated in the example of FIG. **6**, the switch SW.sub.3 connects the first signal port **238** to the first transmission line cable **62** and the switch SW.sub.4 connects the second signal port **240** to the second transmission line cable **64** to facilitate propagation of the transmit and receive signals between the user communication system and the antenna system **50** via the respective transmission line cables **62** and **64**. However, in the calibration mode, such as initiated by the calibration command CAL, the switches SW.sub.3 and SW.sub.4 can be switched to couple each of the transmission line cables **62** and **64** to ground.

As a result, the calibration signal CS.sub.1, such as provided in the example of FIG. **3**, can be provided from the first transmission line cable **62** and can be reflected back to the antenna system **50** as the return signal RTN.sub.1 along the first transmission line cable **62**. Similarly, the calibration signal CS.sub.2, such as provided in the example of FIG. **3**, can be provided from the second transmission line cable **64** and can be reflected back to the antenna system **50** as the return signal RTN.sub.2 along the second transmission line cable **64**. Therefore, with minimal input, the calibration circuit **230** can be implemented in the calibration operation to determine the signal loss of the transmission line cables **62** and **64**.

[0053] FIG. **7** illustrates an example of a transmission line measurement circuit **250**. The transmission line measurement circuit **250** can correspond to the transmission line measurement circuit **68** in the example of FIG. **2**. Therefore, reference is to be made to the examples of FIGS. **2-6** in the following description of the example of FIG. **7**.

[0054] As described previously, the transmission line measurement circuit **250** is configured to determine a signal loss between the user communication system and the antenna circuit **50** through the transmission line cables **62** and **64**. In the example of FIG. **7**, the transmission line measurement circuit **250** includes a calibration signal generator **252** and a signal monitor **254**. The calibration signal generator **252** includes an RF signal source **256** that is configured to generate a calibration signal that can correspond to the first calibration signal CS.sub.1. As an example, the first calibration signal CS.sub.1 can correspond to a dummy RF signal (e.g., a sinusoidal signal) having a predefined frequency. In the example of FIG. **7**, the calibration signal generator **252** includes a resistive network **258** that includes a first resistor R.sub.1 and a second resistor R.sub.2 that are each connected to the RF signal source **256** and to a third resistor R.sub.3 opposite the RF signal source **256** and interconnecting the first and second resistors R.sub.1 and R.sub.2. The resistive network **258** can thus provide a divided version of the first calibration signal CS.sub.1 that can be provided to the first transmission line cable **62** (e.g., via the switch SW.sub.1) in the calibration mode. The transmission line measurement circuit **250** can thus transmit the calibration signal CS.sub.1 to the user communication system **12** via the transmission line cable **62**.

[0055] As described previously, the user communication system **12** can be configured to retransmit the calibration signal CS.sub.1 back to the antenna system **50** from the user communication system **12** as respective return signal RTN.sub.1 on the transmission line cable **64**, such as described in the example of FIG. **4** during the calibration procedure. In the example of FIG. **7**, the return signal RTN.sub.1 is provided to the signal monitor **254** via a diode D.sub.1. The signal monitor **254** is also configured to receive a divided version of the first calibration signal CS.sub.1, demonstrated as a signal CS.sub.1D, via a diode D.sub.2. Therefore, as an example, the signal monitor **254** can be configured to measure a power of each of the return signal RTN.sub.1 and the signal CS.sub.1D. As a result, the signal monitor **254** can calculate a power ratio between the return signal RTN.sub.1 and the signal CS.sub.1D. Accordingly, the signal monitor **254** can determine the signal loss exhibited by the transmission line cables **62** and **64** based on the power ratio between the return signal RTN.sub.1 and the signal CS.sub.1D. The power monitor **254** can thus provide the calculated signal loss, demonstrated as a signal PWR_LS, to the memory **74**.

[0056] While the transmission line measurement circuit **250** is configured to provide the calibration operation demonstrated in the example of FIG. **4**, it is to be understood that the transmission line measurement circuit **250** is not limited to the example of FIG. **7**. For example, the transmission line measurement circuit **250** can be arranged in a manner to facilitate the calibration signal CS.sub.1 and the return signal RTN.sub.1 or the calibration signal CS.sub.2 and the return signal RTN.sub.2 to be propagated along a given one of the transmission line cables **62** and **64**, such as described in the example of FIG. **3**.

[0057] Referring back to the example of FIG. **2**, the antenna circuit **52** also includes a controller **76**. In response to the determining the signal loss, the transmission line measurement circuit **68** can provide the determined signal loss, demonstrated in the example of FIG. **2** as “SM”, to the controller **76** (e.g., from the memory). In response to the determined signal loss SM, the controller **76** can be configured to adjust an amplitude of at least one of the transmit signals transmitted from the antenna system **50** and the receive signals received at the antenna system **50** based on the determined signal loss. As an example, the controller **76** can include a processor that is configured to determine the appropriate adjustments to the amplitude of each of the respective transmit and receive signals based on the determined signal loss SM. While the controller **76** is described as including a processor, the term “processor” can be used to describe other types of processing devices, such as a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), or other type of processing device.

[0058] In the example of FIG. **2**, the antenna circuit **52** includes a first amplitude adjustment circuit **78** that is provided in the first signal path **54** and a second amplitude adjustment circuit **80** that is provided in the second signal path **56**. As an example, the amplitude adjustment circuits **78** and **80** can each include at least one variable circuit element (VCE) in the respective signal paths **54** and **56**. For example, the VCEs can be configured as variable attenuators, variable gain amplifiers, and/or fixed gain amplifiers. In the example of FIG. **2**, the controller **76** is configured to provide control signals, demonstrated as “AT.sub.1” and “AT.sub.2”, respectively, to the amplitude adjustment circuits **78** and **80** based on the determined signal loss SM, as well as based on whether the signal paths **54** and/or **56** are in the transmit mode or the receive mode.

[0059] For example, the communication system **10** can be configured to operate based on a predetermined communication standard that can dictate a predetermined maximum effective isotropic radiated power (EIRP), such as +23 dBm for the transmit signals. Therefore, the controller **76** provides the control signals AT.sub.1 and AT.sub.2 to the respective amplitude adjustment circuits **78** and **80** in the transmit mode to attenuate the transmit signals down to less than the predetermined maximum EIRP. For example, the antenna arrays **58** and **60** can be designed with sufficiently high gain to provide the transmit signals at a power level that is greater than the predetermined maximum EIRP to overcome power losses of the transmission line cables **62** and **64** regardless of the length of the transmission line cables **62** and **64**, such that the transmit signals can be attenuated down to approximately the predetermined maximum EIRP. Additionally or alternatively, the signal paths **54** and **56** can include sufficient power amplification in the transmit mode, as described in greater detail herein, to overcome power losses of the transmission line cables **62** and **64** regardless of the length of the transmission line cables **62** and **64**, such that the transmit signals can be attenuated down to approximately the predetermined maximum EIRP. Similarly, the controller **76** provides the control signals AT.sub.1 and AT.sub.2 to the respective amplitude adjustment circuits **78** and **80** in the receive mode to attenuate the receive signals down to less than an acceptable operational level (e.g., a maximum saturation power associated with the antenna circuit **50** and/or the user communication system **12**). Therefore, the antenna system **50** can be installed in a manner that is substantially agnostic of the length and/or loss characteristics of the transmission line cables **62** and **64** based on the calibration operation to determine the signal loss of the transmission line cables **62** and **64**.

[0060] As described previously, the communication system **10** can operate based on a TDD communication standard, such that the transmit signals and the receive signals can be interleaved with each other on a given signal path between the user communication system **12** and the antenna arrays **58** and **60**. In addition, as described previously, the signal paths **54** and **56** of the antenna circuit **52** can operate in either a transmit mode or a receive mode corresponding to transmission of the transmit signals or receipt of the receive signals in the TDD manner along the respective signal paths **54** and **56**. In the example of FIG. **2**, the antenna circuit **52** also includes a first transmit detection circuit **90** associated with the first signal path **54** and a second transmit detection circuit **92** associated with the second signal path **56**. The transmit detection circuits **90** and **92** can be configured to measure power on the respective signal paths **54** and **56** to determine if the user communication system **12** is transmitting a transmit signal.

[0061] For example, the transmit detection circuits **90** and **92** can each include a bi-directional coupler with a terminated load to determine if the power on the respective one of the signal paths **54** and **56** is greater than a predetermined threshold to determine if the user communication system **12** is transmitting a transmit signal. In the example of FIG. **2**, the transmit detection circuits **90** and **92** are configured to generate mode signals TX.sub.1 and TX.sub.2 that are provided to the controller **76**, such as to indicate that the respective one of the signal paths **54** and **56** is in the transmit mode. Therefore, in response to the transmit detection circuits **90** and **92** determining if the user communication system **12** is transmitting a transmit signal, the controller **76** can switch the respective signal paths **54** and **56** from the receive mode as a default mode to a transmit mode to facilitate transmission of the transmit signal from the antenna system **50** via the antenna arrays **58** and **60**. Similarly, in response to the transmit detection circuits **90** and **92** detecting a decrease in the power of the signal path (e.g., less than the predetermined threshold), the controller **76** can switch the respective signal paths **54** and **56** back to the receive mode from the transmit mode (e.g., upon expiration of a timer).
[0062] As described previously, the controller **76** can be configured to adjust the respective control signal(s) AT.sub.1 and AT.sub.2 based on the indication of the transmit mode or the receive mode of the respective one of the signal paths **54** and **56**, such as based on the respective mode signals TX.sub.1 and TX.sub.2. Therefore, the amplitude of the transmit and receive signals can be adjusted (e.g., attenuated) based on whether the respective signal path **54** or **56** is in the transmit mode or the receive mode. As another example, as described previously, the amplitude adjustment circuits **78** and **80** can switch between a transmit mode signal path and a receive mode signal path for each of the amplitude adjustment circuits **78** and **80** via switches. Therefore, the controller **76** can also include a switch controller **94** that is configured to control the switches of the amplitude adjustment circuits **78** and **80**.
[0063] As an example, the switch controller **94** can be configured to generate mode signals MD.sub.1 and MD.sub.2, respectively, to control the mode of the respective one of the signal paths **54** and **56**. For example, the amplitude adjustment circuit **78** can be controlled by the first switching signal MD.sub.1 and the amplitude adjustment circuit **80** can be controlled by the second switching signal MD.sub.2. In response to one of the transmit detection circuits **90** and **92** determining that the user communication system **12** is transmitting a transmit signal along the respective one of the signal paths **54** and **56**, the respective one of the transmit detection circuits **90** and **92** commands the controller **76** (e.g., via the mode signals TX.sub.1 and TX.sub.2) to provide the respective one of the switching signals MD.sub.1 and MD.sub.2 to the respective one of the amplitude adjustment circuits **78** and **80**. In response to the respective one of the switching signals MD.sub.1 and MD.sub.2, the respective amplitude adjustment circuit **78** and **80** can activate at least one switch to switch the respective amplitude adjustment circuit **78** or **80** from the default receive mode to the transmit mode to facilitate transmission of the transmit signal along the respective signal path **54** and **56** and from the respective antenna array **58** and **60**.
[0064] FIG. **8** illustrates an example of a controller **300**. The controller **300** can correspond to the controller **76** in the example of FIG. **2**. Therefore, reference is to be made to the example of FIG. **2** in the following description of the example of FIG. **8**.
[0065] The controller **300** includes a processor **302**. For example, the processor **302** can communicate with or include the amplitude adjustment circuits **78** and **80**. In the example of FIG. **8**, the processor **302** receives the signal SM corresponding to the signal loss of the transmission line cables **62** and **64**. Therefore, the processor **302** can be configured to calculate the appropriate amount of amplification (e.g., attenuation) to provide to the amplitude adjustment circuits **78** and **80**, such as to provide the appropriate attenuation to the signal paths **54** and/or **56** based on the respective mode (e.g., transmit mode or receive mode). In the example of FIG. **7**, the processor **302** is demonstrated as generating the control signals AT.sub.1 and AT.sub.2 that are provided to the amplitude adjustment circuits **78** and **80**, such as to provide the appropriate attenuation to the signal

paths **54** and/or **56** based on the respective mode.

[0066] As described previously, the controller **76** in the example of FIG. **2** can include the switching controller **94** to control the switches of the amplitude adjustment circuits **78** and **80**. In the example of FIG. **8**, the controller **300** can include a switching controller **304** that can provide a switching signal MD for one of the amplitude adjustment circuits **78** and **80**. Therefore, it is to be understood that the controller **300** can include a switching controller **304** for each of the signal paths **54** and **56**. The switching controller **304** includes a first comparator **306** and a second comparator **308**. The first comparator **306** is configured to compare a voltage V.sub.RX corresponding to an approximate power of the receive signals with a threshold voltage V.sub.RX_TH. Similarly, the second comparator **308** is configured to compare a voltage V.sub.TX corresponding to an approximate power of the transmit signals with a threshold voltage V.sub.TX_TH. As an example, the voltages V.sub.RX and V.sub.TX can correspond to the same voltage (e.g., corresponding to the signal power on a given one of the signal paths **54** and **56**, as measured by the transmit detection system(s) **90** and **92**. Therefore, the comparators **306** and **308** can be configured to provide an asserted output corresponding to the mode of the respective signal path **54** or **56**.

[0067] The switching controller **304** includes a first sequence of D latches (e.g., flip-flops), demonstrated as **310**, **312**, and **314**. The first D latch **310** receives the output of the first comparator **306** as an input, with the D latches **310**, **312**, and **314** being configured in a cascaded arrangement from output to input. Each of the D latches **310**, **312**, and **314** receives a clock signal CLK from an oscillator **316**. The output of the second D latch **312** and the third D latch **314** are provided as inputs to an AND-gate **318**, with the input received from the third D latch **314** being inverted. In a similar arrangement, the switching controller also includes a second sequence of D latches, demonstrated as **320**, **322**, and **324**. The first D latch **320** receives the output of the second comparator **308** as an input, with the D latches **320**, **322**, and **324** being configured in a cascaded arrangement from output to input. Each of the D latches **320**, **322**, and **324** likewise receives the clock signal CLK. The output of the second D latch **322** and the third D latch **324** are provided as inputs to an AND-gate **326**, with the input received from the third D latch **324** being inverted.

[0068] The output of the AND-gate **318** is provided as a set input to an SR latch **328**, and the output of the AND-gate **326** is provided as a reset input to the SR latch **328**. The SR latch **328** likewise the clock signal CLK, and is configured to generate the respective switching signal MD (e.g., one of the switching signals MD.sub.1 and MD.sub.2). Therefore, the SR latch **328** is configured to rapidly change the state of the switching signal MD in response to a change in amplitude of the voltages V.sub.TX and/or V.sub.RX. For example, in response to a change of the amplitude of the voltages V.sub.TX and/or V.sub.RX, the logic sequence of the D latches **310**, **312**, **314**, **320**, **322**, and **324**, the AND-gates **318** and **326**, and the SR latch **328** can be configured to change the state of the switching signal MD in a time of approximately 10 microseconds or less, such as to satisfy the TDD communication standard.

[0069] The processor **302** can be configured to receive a plurality of inputs associated with the switching logic of the switching controller **304**. In the example of FIG. **8**, the processor **302** receives as inputs the output of the D latches **314** and **324**, as well as the outputs of the AND-gates **318** and **326**. For example, the processor **302** can be configured as a state machine to monitor the state of the signal path (e.g., the signal path **54** or **56**), such that the inputs to the inputs to the processor **302** are configured to set flags and/or registers for operation of the antenna circuit **52**. As another example, the oscillator **316** can be included in the processor **302**, such that the processor **302** generates the clock signal CLK. Additionally, the processor **302** is demonstrated as generating the predetermined threshold voltages V.sub.TX_TH and V.sub.RX_TH, which can be programmable via input to the processor **302**, or can have fixed voltage amplitudes.

[0070] In the example of FIG. **8**, the processor **302** includes timers **330** (e.g., one for each of the signal paths **54** and **56**). As an example, the timers **330** can correspond to watchdog timers for controlling timing associated with the mode selection of the respective signal paths **54** and **56**. For example, in response to the inputs provided to the processor **302** indicating that the mode is set for transmit mode for a given one of the signal path(s) **54** and **56**, but the transmit power is less than the predetermine threshold (e.g., the voltage V.sub.TX is less than the predetermined threshold V.sub.TX_TH), the respective timer **330** can begin counting a predetermined timing threshold. As an example, in response to the respective timer **330** counting for a predetermined time duration (e.g., approximately one millisecond), the processor **302** can switch back to a default receive mode for the given signal path **54** or **56**, such as to change the amplitude of the respective one of the control signals AT.sub.1 and AT.sub.2. Additionally, the processor **302** can assert an output to the SR latch **328** (e.g., to a “clear” input of the SR latch **328**). Therefore, the SR latch **328** can reset to change the state of the switching signal MD to indicate switching the mode from the transmit mode back to the receive mode. As a result, the signal path(s) **54** and/or **56** can be returned to the default receive mode in response to the timed indication of no more transmit signals being transmitted from the user communication system **12**.

[0071] In order to satisfy a given TDD communication standard, the control signals AT.sub.1 and AT.sub.2 and the amplitude adjustment circuits **78** and **80** may be required to switch between the transmit mode and the receive mode as quickly as possible. FIG. **9** illustrates an example diagram **350** of a TDD communication stream. The TDD communication stream includes a first set of receive signal sub-frames, demonstrated at **352**, a first set of transmit signal sub-frames, demonstrated at **354**, and a second set of receive mode sub-frames **356**. As an example, the TDD communication stream can continue thereafter with alternating sets of transmit signal sub-frames and receive signa sub-frames in a TDD manner. The transmit and receive signal sub-frames are demonstrated in the example of FIG. **9** as being demonstrated as a function of time. While each of the transmit and receive signal sub-frames are demonstrated as being approximately equal in time, it is to be understood that the transmit and receive signal sub-frames are not necessarily equal in length of time, and that the elements of the time domain demonstrated in the example of FIG. **9** are not necessarily illustrated to scale.

[0072] In the example of FIG. **9**, between each of the sets of receive signal sub-frames (e.g., the receive signal sub-frames **352**) and transmit signal sub-frames (e.g., the transmit signal sub-frames **354**) is a time TINT. The time TINT can, for example, correspond to a substantial maximum intermediate time between propagation of the receive signal sub-frames and the transmit signal sub-frames on a given signal path, such as between the user communication system **12**, along a transmission line cable **62** or **64**, along a respective signal path **54** or **56** in the antenna circuit **52**, and a respective one of the antenna arrays **58** or **60**, such as defined by the predetermined TDD communication standard.

[0073] In the example of FIG. **9**, the time TINT between each of the transmit and receive signal sub-frames includes a first portion of time **358** and a second portion of time **360**. The first portion of time **358** can correspond to a switching time (e.g., approximately 10 microseconds or less), such as to generate the appropriate switching signal(s) MD.sub.1 and MD.sub.2 and/or to activate the respective switches of the amplitude adjustment circuits **78** and **80**. The second portion of time **360** can correspond to switch settling time (e.g., likewise approximately 10 microseconds or less), such as a time for the respective switches to settle to a saturation region and/or to dissipate parasitic effects (e.g., capacitance and/or inductance) of the circuit components of the switching controller **304** and/or the respective amplitude adjustment circuits **78** and **80**. Therefore, the hardware-based logic circuit of the switching controller **304** can implement rapid state-changes of the switching signal(s) MD.sub.1 and MD.sub.2 to satisfy the rapid switching requirements dictated by the TDD communication standard.

[0074] FIGS. **10-15** demonstrate examples of amplitude adjustment circuits. The example of FIG. **10** demonstrates an amplitude adjustment circuit **370**, the example of FIG. **11** demonstrates an amplitude adjustment circuit **400**, the example of FIG. **12** demonstrates an amplitude adjustment circuit **450**, the example of FIG. **13** demonstrates an amplitude adjustment circuit **500**, the example of FIG. **14** demonstrates an amplitude adjustment circuit **550**, and the example of FIG. **15** demonstrates an amplitude adjustment circuit **600**. Any of the amplitude adjustment circuits **370**, **400**, **450**, **500**, **550**, and **600** can correspond to the amplitude adjustment circuits **78** and **80** in the example of FIG. **2**. Therefore, reference is to be made to the example of FIG. **2** in the following description of the examples of FIGS. **10-14**. Additionally, the amplitude adjustment circuits **370**, **400**, **450**, **500**, **550**, and **600** are not limited to the examples demonstrated in the examples of FIGS. **10-14**. For example, the amplitude adjustment circuits **370**, **400**, **450**, **500**, **550**, and **600** can include filters, such as low-noise filters, bandpass filters, and the like that can be arranged in the respective transmit path, receive path, or both. Furthermore, the switches described in the amplitude adjustment circuits **400**, **450**, **500**, **550**, and **600** can be implemented as transistor devices, such as to provide very rapid switching times between the transmit mode and the receive mode.

[0075] In the example of FIG. **10**, the amplitude adjustment circuit **370** includes a VCE **372** in the signal path (e.g., the signal path **54** or **56**). The VCE **372** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **372** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). Therefore, the mode of the amplitude adjustment circuit **370** is controlled by the amount of adjustment (e.g., attenuation) provided by the control signal AT in each of the transmit mode and the receive mode.

[0076] In the example of FIG. **11**, the amplitude adjustment circuit **400** includes a VCE **401** in the signal path (e.g., the signal path **54** or **56**). The VCE **401** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **401** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). The amplitude adjustment circuit **400** also includes first switch SW.sub.5, a second switch SW.sub.6, and a third switch SW.sub.7 that are each controlled by the switching signal MD. The switches SW.sub.5, SW.sub.6, and SW.sub.7 are demonstrated in a default state corresponding to the default state of the receive mode. The first and second switches SW.sub.5 and SW.sub.6 are each demonstrated in the example of FIG. **11** as single-pole double-throw switches that select between a first signal path, demonstrated at **402**, and a second signal path, demonstrated at **404**. In the example of FIG. **11**, the first signal path **402** can correspond to the receive mode and the second signal path **404** can correspond to the transmit mode. The second signal path **404** includes a power amplifier **406** that is configured to amplify the transmit signal in the transmit mode. Additionally, the amplitude adjustment circuit **400** includes a low-noise amplifier (LNA) **408** that is arranged in parallel with the third switch SW.sub.7, arranged as a single-pole single-throw switch. Therefore, in the receive mode, the receive signal is amplified by the LNA **408**, and in the transmit mode, the transmit signal is provided in a bypass short-circuit through the closed switch SW.sub.7.

[0077] In the example of FIG. **12**, the amplitude adjustment circuit **450** includes a VCE **451** in the signal path (e.g., the signal path **54** or **56**). The VCE **451** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **451** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). The amplitude adjustment circuit **450** also includes a first circulator **452**, a second circulator **454**, a first switch SW.sub.5, and a second switch SW.sub.6 that are each controlled by the switching signal MD. The switches SW.sub.5 and SW.sub.6 are demonstrated in a default state corresponding to the default state of the receive mode. The first and second switches SW.sub.5 and SW.sub.6 are each demonstrated in the example of FIG. **11** as single-pole double-throw switches. The first circulator **452** is demonstrated as a "clockwise" circulator, such that the transmit signal is provided to the first switch SW.sub.5. The first switch SW.sub.5 selects between an attenuator **456** in the receive mode and a power amplifier **458** in the transmit mode, and thus is output from the amplitude adjustment circuit **450** via the second circulator **454** that is also arranged as a "clockwise" circulator. The second circulator **454** also provides the receive signal to the second switch SW.sub.6. The second switch SW.sub.6 selects between an attenuator **460** in the transmit mode and an LNA **462** in the receive mode, and thus is output from the amplitude adjustment circuit **450** via the first circulator **452**.

[0078] In the example of FIG. **13**, the amplitude adjustment circuit **500** includes a VCE **501** in the signal path (e.g., the signal path **54** or **56**). The VCE **501** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **501** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). The amplitude adjustment circuit **500** also includes a switch SW.sub.5 and a circulator **502**. The switch SW.sub.5 is arranged as a single-pole double-throw switch controlled by the switching signal MD, and is demonstrated in a default state corresponding to the default state of the receive mode. The switch SW.sub.5 selects between a first signal path, demonstrated at **504**, and a second signal path, demonstrated at **506**. In the example of FIG. **13**, the first signal path **504** can correspond to the transmit mode and the second signal path **506** can correspond to the receive mode. The first signal path **504** includes a power amplifier **508** that is configured to amplify the transmit signal in the transmit mode, and thus is output from the amplitude adjustment circuit **500** via the circulator **502** that is also arranged as a "clockwise" circulator. The second signal path **506** includes an LNA **510**, such that the circulator **502** provides the receive signal on the second signal path **506** to be amplified by the LNA **510** and output from the amplitude adjustment circuit **500** via the switch SW.sub.5 in the receive mode.

[0079] In the example of FIG. **14**, the amplitude adjustment circuit **550** includes a VCE **551** in the signal path (e.g., the signal path **54** or **56**). The VCE **551** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **551** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). The amplitude adjustment circuit **550** also includes a switch SW.sub.5 and a circulator **552**. The switch SW.sub.5 is arranged as a single-pole double-throw switch controlled by the switching signal MD, and is demonstrated in a default state corresponding to the default state of the receive mode. The switch SW.sub.5 selects between a first signal path, demonstrated at **554**, and a second signal path, demonstrated at **556**. In the example of FIG. **14**, the first signal path **554** can correspond to the transmit mode and the second signal path **556** can correspond to the receive mode. The first signal path **554** is demonstrated as a bypass short-circuit to output the transmit signal from the amplitude adjustment circuit **550** via the circulator **552** that is also arranged as a "clockwise" circulator. The second signal path **556** includes an LNA **558**, such that the circulator **552** provides the receive signal on the second signal path **556** to be amplified by the LNA **558** and output from the amplitude adjustment circuit **550** via the switch SW.sub.5 in the receive mode.

[0080] In the example of FIG. **15**, the amplitude adjustment circuit **600** includes a VCE **601** in the signal path (e.g., the signal path **54** or **56**). The VCE **601** is demonstrated as being controlled by a control signal AT (e.g., one of the control signals AT.sub.1 or AT.sub.2). For example, the VCE **601** can be configured as a variable attenuator that is controlled by the controller **76** to provide attenuation of the transmit signals in the transmit mode and to provide attenuation of the receive

signals in the receive mode (e.g., based on a respective one of the mode signals TX.sub.1 or TX.sub.2). The amplitude adjustment circuit **600** also includes a switch SW.sub.5 controlled by the switching signal MD. The switch SW.sub.5 is demonstrated as a single-pole single-throw switch in a default state corresponding to the default state of the receive mode. The amplitude adjustment circuit **600** includes an LNA **602** that is arranged in parallel with the switch SW.sub.5. Therefore, in the receive mode, the receive signal is amplified by the LNA **602**, and in the transmit mode, the transmit signal is provided in a bypass short-circuit through the closed switch SW.sub.5.
[0081] The examples of FIGS. **10**, **14**, and **15** do not include power amplifiers to provide amplification of the transmit signals in the transmit mode. As described previously, the signal paths **54** and **56** can include sufficient power amplification in the transmit mode, such as provided in the examples of FIGS. **11-13**, to overcome power losses of the transmission line cables **62** and **64** regardless of the length of the transmission line cables **62** and **64** (e.g., to attenuate the transmit signals down to approximately the predetermined maximum EIRP). As another example, the amplitude adjustment circuits **370**, **550**, and **600** in the examples of FIGS. **10**, **14**, and **15**, respectively, can be implemented when the user communication system **12** includes sufficient power amplification of the transmit signals that power amplifiers are not necessary in the transmit signal paths. Additionally or alternatively, the antenna arrays **58** and **60** can be designed with sufficiently high gain that power amplification of the transmit signals is not necessary in the transmit signal paths to provide feasibility of the amplitude adjustment circuits **550** and **600**.
[0082] As another example, the amplitude adjustment circuits **400**, **450**, and **500** can be implemented for installation of an antenna system **50** in a manner that is completely agnostic of the user communication system **12**. For example, during the calibration procedure, in addition to measuring the signal loss of the transmission line cables **62** and **64**, the antenna system **50** can measure an output power of the transmit signals provided from the user communication system **12** (e.g., via the transmit detection circuits **90** and **92**, such as relative to a plurality of thresholds). Therefore, in response to determining the output power of the user communication system **12**, the antenna circuit **52** can properly attenuate the transmit signals in the transmit mode down to approximately the predetermined maximum EIRP.
[0083] As a result, the switching controller **304** can implement state changes of the switching signal MD in response to changes of amplitude of the voltages V.sub.TX and/or V.sub.RX, such as in response to the transmit detection circuit(s) **90** and **92** detecting changes in power on the respective signal path(s) **54** and **56**. Accordingly, the switching signals MD.sub.1 and MD.sub.2 can provide sufficiently rapid switching to satisfy the maximum switching times (e.g., the first portion of time **558** in the example of FIG. **9**) to comply with the TDD communication standard. As a result, the antenna system **50** can operate to facilitate the bidirectional TDD communications between transmit and receive signals, such as without requiring communication or signal transfer from the user communication system **12**. Therefore, the antenna system **50** can be installed in a simplistic manner that is largely independent of the operation of the user communication system **12**. Additionally, the antenna system **14** can be installed in a manner that is agnostic of the length of the transmission line cable(s) **62** and **64** interconnecting the antenna system **50** and the user communication system **12**. Accordingly, the antenna system **50** can be simplistically installed to efficiently facilitate wireless communication between the user communication system **12** and a network hub (e.g., a base station).
[0084] In view of the foregoing structural and functional features described above, a methodology in accordance with various aspects of the present invention will be better appreciated with reference to FIG. **16**. While, for purposes of simplicity of explanation, the methodology of FIG. **16** is shown and described as executing serially, it is to be understood and appreciated that the present invention is not limited by the illustrated order, as some aspects could, in accordance with the present invention, occur in different orders and/or concurrently with other aspects from that shown and described herein. Moreover, not all illustrated features may be required to implement a

methodology in accordance with an aspect of the present invention.
[0085] FIG. **16** illustrates an example of a method **650** for communicating at least one of a transmit signal and a receive signal via a TDD antenna system (e.g., the antenna system **14**) comprising an antenna (e.g., the antenna array(s) **18**). At **652**, a calibration signal (e.g., the calibration signal(s) CS.sub.1 and/or CS.sub.2) is provided from an antenna circuit (e.g., the antenna circuit **20**) associated with the antenna system to a user communication system (e.g., the user communication system **12**) on at least one transmission line cable (e.g., the transmission line cable(s) **16**). At **654**, a returned signal (e.g., the return signal(s) RTN.sub.1 and RTN.sub.2) corresponding to the calibration signal retransmitted back from the user communication system is received at the antenna circuit. At **656**, signal loss between the user communication system and the antenna circuit through at least one transmission line cable is determined (e.g., via the transmission line measurement circuit **22**) based on the returned signal. At **658**, signal power of the transmit signal obtained from the user communication system via the at least one transmission line cable is monitored (e.g., via the transmit detection circuit **26**). At **660**, an amplitude of the receive signal is adjusted in a receive mode based on the determined signal loss. At **662**, an amplitude adjustment circuit (e.g., the amplitude adjustment circuit **24**) is switched from the receive mode to a transmit mode (e.g., via the controller **28**) in response to the monitored signal power exceeding a predetermined threshold. At **664**, an amplitude of the transmit signal is adjusted in the transmit mode based on the determined signal loss.
[0086] What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Additionally, where the disclosure or claims recite “a,” “an,” “a first,” or “another” element, or the equivalent thereof, it should be interpreted to include one or more than one such element, neither requiring nor excluding two or more such elements. As used herein, the term “includes” means includes but not limited to, and the term “including” means including but not limited to. The term “based on” means based at least in part on.

Claims

1. A self-synchronizing time-division duplex (TDD) antenna system comprising: an antenna configured to communicate a transmit signal and a receive signal; and an antenna circuit coupled to the antenna, and for coupling to a user communication system via at least one transmission line cable, the antenna circuit comprising: a transmission line measurement circuit configured to determine signal loss between the user communication system and the antenna circuit through the at least one transmission line cable, wherein the transmission line measurement circuit comprises: a calibration signal generator configured to provide a calibration signal on the at least one transmission line cable from the antenna circuit to the user communication system; and a signal monitor configured to: receive a return calibration signal transmitted back from the user communication system as a returned signal; and determine the signal loss based on a characteristic of the returned signal; and an amplitude adjustment circuit configured to adjust amplitude of at least one of the transmit and receive signals based on the determined signal loss.
2. The system of claim 1, wherein the antenna circuit further comprises: a transmit detection circuit configured to monitor signal power of the transmit signal obtained from the user communication system via the at least one transmission line cable; and a controller configured to switch the amplitude adjustment circuit from a receive mode to a transmit mode in response to the monitored signal power exceeding a predetermined threshold, wherein in the receive mode the amplitude

adjustment circuit is configured to apply a receive amplitude adjustment along a signal path to the receive signal, and in the transmit mode the amplitude adjustment circuit is configured to apply a transmit amplitude adjustment along the signal path to the transmit signal.

3. The system of claim 2, wherein in the receive mode the amplitude adjustment circuit is configured to adjust the amplitude of the receive signal obtained via the antenna and is configured to provide the adjusted receive signal to the user communication system via the at least one transmission line cable, and in the transmit mode the amplitude adjustment circuit is configured to adjust the amplitude of the transmit signal.

4. The system of claim 1, wherein the calibration signal generator provides the calibration signal in response to a calibration command.

5. The system of claim 1, wherein the signal monitor is configured to measure a first power associated with the calibration signal provided on the at least one transmission line cable and to measure a second power associated with the returned signal on the at least one transmission line cable to determine the signal loss as a ratio of the first power and the second power.

6. The system of claim 5, wherein the at least one transmission line cable comprises a first transmission line cable and a second transmission line cable, wherein the signal monitor is configured to measure the first power associated with the calibration signal provided on the first transmission line cable and to measure the second power associated with the returned signal received on the second transmission line cable.

7. The system of claim 1, wherein the antenna system comprises an extraction circuit configured to receive DC power from the user communication system via the at least one transmission line cable.

8. The system of claim 2, wherein the controller is further configured to switch from the transmit mode to the receive mode in response to the monitored signal power being below the predetermined threshold.

9. The system of claim 2, wherein: the transmit signal is a first transmit signal and the receive signal is a first receive signal; the antenna comprises a first antenna array to communicate the first transmit signal and the first receive signal, and further comprises a second antenna array to communicate a second transmit signal and a second receive signal; the signal path of the amplitude adjustment circuit is a first signal path; and in the receive mode the amplitude adjustment circuit is configured to apply the receive amplitude adjustment along a second signal path to the second receive signal, and in the transmit mode the amplitude adjustment circuit is configured to apply the transmit amplitude adjustment along the second signal path to the second transmit signal.

10. The system of claim 9, wherein: the at least one transmission line cable includes a first transmission line cable and a second transmission line cable; and the antenna circuit communicates the first receive signal and the first transmit signal with the user communication system via the first transmission line cable, and communicates the second receive signal and the second transmit signal with the user communication system via the second transmission line cable.

11. The system of claim 1, wherein the antenna circuit further comprises memory to store the determined signal loss.

12. The system of claim 2, wherein the transmit detection circuit comprises a directional coupler and power detector to monitor the signal power of the transmit signal.

13. A method for communicating at least one of a transmit signal and a receive signal via a time-division duplex (TDD) antenna system comprising an antenna, the method comprising: providing a calibration signal from an antenna circuit associated with the antenna system to a user communication system on at least one transmission line cable; receiving a returned signal corresponding to the calibration signal retransmitted back from the user communication system to the antenna circuit; determining signal loss between the user communication system and the antenna circuit through at least one transmission line cable based on the returned signal; and adjusting an amplitude of at least one of the transmit and receive signals based on the determined signal loss.

14. The method of claim 13, wherein determining signal loss comprises: detecting a characteristic of the returned signal; and determining the signal loss based on the characteristic of the returned signal.
15. The method of claim 13, wherein providing the calibration signal comprises providing the calibration signal in response to a calibration command.
16. The method of claim 13, wherein determining signal loss comprises: measuring a first power associated with the calibration signal provided on the at least one transmission line cable; and measuring a second power associated with the returned signal on the at least one transmission line cable to determine the signal loss as a ratio of the first power and the second power.
17. The method of claim 16, wherein the at least one transmission line cable comprises a first transmission line cable and a second transmission line cable, wherein measuring the first power comprises measuring the first power associated with the calibration signal provided on the first transmission line cable and measuring the second power comprises measuring the second power associated with the returned signal received on the second transmission line cable.
18. The method of claim 13, wherein adjusting the amplitude of at least one of the transmit and receive signals comprises: adjusting the amplitude of the receive signal in a receive mode; and adjusting the amplitude of the transmit signal in a transmit mode.
19. The method of claim 18, further comprising: monitoring signal power of the transmit signal obtained from the user communication system via the at least one transmission line cable; and switching an amplitude adjustment circuit between the receive mode and the transmit mode based on the monitored signal power and a predetermined threshold.
20. The method of claim 18, wherein: the transmit signal is a first transmit signal and the receive signal is a first receive signal; the antenna comprises a first antenna array to communicate the first transmit signal and the first receive signal, and further comprises a second antenna array to communicate a second transmit signal and a second receive signal; adjusting the amplitude of the receive signal in the receive mode comprises adjusting the amplitude of a second receive signal in a second receive path based on the determined signal loss; and adjusting the amplitude of the transmit signal in the transmit mode comprises adjusting the amplitude of a second transmit signal in a second transmit path based on the determined signal loss.
21. The method of claim 20, wherein the at least one transmission line cable comprises: a first transmission line cable on which the first receive signal and the first transmit signal are communicated between the user communication system and the antenna system; and a second transmission line cable on which the second receive signal and the second transmit signal are communicated between the user communication system and the antenna system.
22. The method of claim 13, further comprising storing the determined signal loss in a memory.
23. The method of claim 19, wherein monitoring the signal power of the transmit signal comprises monitoring the signal power of the transmit signal via a directional coupler and a power detector.